

LESSON - 17

TRIGONOMETRY

Angles	18°	36°	54°	72°
Sin	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10+2\sqrt{5}}}{4}$
cos	$\frac{\sqrt{10+2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}-1}{4}$
θ	15°	$22\frac{1}{2}^\circ$	$67\frac{1}{2}^\circ$	75°
Sin θ	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$	$\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$	$\frac{\sqrt{3}+1}{2\sqrt{2}}$
cos θ	$\frac{\sqrt{3}+1}{2\sqrt{2}}$	$\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$	$\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$	$\frac{\sqrt{3}-1}{2\sqrt{2}}$
Tan θ	$2-\sqrt{3}$	$\sqrt{2}-1$	$\sqrt{2}+1$	$2+\sqrt{3}$
Cot θ	$2+\sqrt{3}$	$\sqrt{2}+1$	$\sqrt{2}-1$	$2-\sqrt{3}$

1. For $A+B+C=180^\circ$

(i) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(ii) $\sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C$

(iii) $\cos 2A + \cos 2B + \cos 2C = 1 - 4 \cos A \cos B \cos C$

(iv) $\cos 2A + \cos 2B - \cos 2C = 1 - 4 \sin A \sin B \cos C$

(v) $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$

(vi) $\sin A - \sin B + \sin C = 4 \sin \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$

(vii) $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

(viii) $\cos A + \cos B - \cos C = -1 + 4 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}$

$$\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\cot 2\theta = \frac{\cot^2\theta - 1}{2\cot\theta}$$

$$(i) \sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$(ii) \cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$(iii) \tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$$

Transformations:

$$(i) \sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$(ii) \sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$$

$$(iii) \cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$(iv) \cos C - \cos D = 2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{D-C}{2}\right)$$

Angles are in AP:

$$\bullet \sin A + \sin(A+D) + \dots + \sin(A+(n-1)D)$$

$$= \sin\left(A + (n-1)\frac{D}{2}\right)\sin\left(\frac{nD}{2}\right)$$

$$\sin\frac{D}{2}$$

$$\bullet \cos A + \cos(A+D) + \dots + \cos(A+(n-1)D)$$

$$= \cos\left(A + (n-1)\frac{D}{2}\right)\sin\left(\frac{nD}{2}\right)$$

$$\sin\left(\frac{D}{2}\right)$$

$$\boxed{AM \geq GM}$$

2. Range of ax^2+bx+c :

3. For $a\cos x + b\sin x = c$

$$(i) \text{ if } a > 0 \text{ Range} = \left[-\frac{D}{4a}, \infty\right)$$

$$\text{Max: } c - \sqrt{a^2+b^2}$$

$$(ii) \text{ if } a < 0 \text{ Range} = \left(-\infty, \frac{-D}{4a}\right]$$

$$\text{Min: } c + \sqrt{a^2+b^2}$$

4. $A+B+C = \pi$.

(i) $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$

(ii) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(iii) $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

(iv) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

(v) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(vi) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

(vii) $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$

5.

(i) $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$

(ii) $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$

(iii) $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$

(iv) $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

6.

(i) $\sin \theta \cdot \sin(60-\theta) \cdot \sin(60+\theta) = \frac{1}{4} \sin 3\theta$

(ii) $\cos \theta \cdot \cos(60-\theta) \cdot \cos(60+\theta) = \frac{1}{4} \cos 3\theta$

(iii) $\tan \theta \cdot \tan(60-\theta) \cdot \tan(60+\theta) = \tan 3\theta$

(iv) $\cot \theta \cdot \cot(60-\theta) \cdot \cot(60+\theta) = \cot 3\theta$

7.

(i) $\tan\left(\frac{\pi}{4} + A\right) = \cot\left(\frac{\pi}{4} + (-A)\right) = \frac{1 + \tan \theta}{1 - \tan \theta} = \frac{\cot \theta + 1}{\cot \theta - 1} = \frac{\cos A + \sin A}{\cos A - \sin A}$

(ii) $\tan\left(\frac{\pi}{4} - \theta\right) = \cot\left(\frac{\pi}{4} + \theta\right) = \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{\cot A - 1}{\cot A + 1} = \frac{\cos A - \sin A}{\cos A + \sin A}$

Important :

$$1. \cos x \cdot \cos 2x \cdot \cos 4x \cdots \cos 2^{n-1}x = \frac{\sin 2^n x}{2^n \sin x}$$

$$2. \cot \theta - \tan \theta = 2 \cot 2\theta$$

$$3. \cos^3 \theta + \cos^3 (120^\circ - \theta) + \cos^3 (120^\circ + \theta) = \frac{3}{4} \cos 3\theta$$

$$4. \sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots \sqrt{2 + 2 \cos \theta}}}} = 2 \cos \left(\frac{\theta}{2^n} \right)$$

$$5. \sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots}}} = 2 \cos \left(\frac{\pi}{2^{n+1}} \right)$$

$$6. (1 + \sec 2\theta)(1 + \sec 2^2\theta) \cdots (1 + \sec 2^n\theta) = \frac{\tan 2^n\theta}{\tan \theta}$$

$$7. \tan A + 2 \tan 2A + \cdots + 2^{n-1} \tan 2^{n-1}A + 2^n \cot 2^n A = \cot A$$

$$8. \sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta = 1 - \frac{1}{2} \sin^2 2\theta$$

$$9. \sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$$

$$= 1 - \frac{3}{4} \sin^2 2\theta$$

$$10. \left(\frac{\cos A + \cos B}{\sin A + (-\sin B)} \right)^n + \left(\frac{\sin A + \sin B}{\cos A - \cos B} \right)^n =$$

$$\text{if } n \text{ is Even} = 2 \cot^n \left(\frac{A+B}{2} \right)$$

$$n \text{ is Odd} = 0$$

$$11. \cos x \cdot \cos 2x \cdot \cos 4x \cdots \cos 2^{n-1}x = \frac{\sin 2^n x}{2^n \sin x}$$

$$12. (i) \sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \cdots + \sin\left[\alpha + (n-1)\beta\right]$$

$$= \frac{\sin \frac{n\beta}{2}}{\sin \frac{\beta}{2}} \sin \left\{ \alpha + \left(\frac{n-1}{2} \right) \beta \right\}$$

$$(ii) \cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \cdots + \cos(\alpha + (n-1)\beta)$$

$$= \frac{\sin \frac{n\beta}{2}}{\sin \frac{\beta}{2}} \cos \left\{ \alpha + \left(\frac{n-1}{2} \right) \beta \right\}$$

$$13. \quad \tan x + \frac{1}{2} \tan \frac{x}{2} + \frac{1}{2^2} \tan \frac{x}{2^2} + \dots + \frac{1}{2^{n-1}} \tan \left(\frac{x}{2^{n-1}} \right)$$

$$= \frac{1}{2^{n-1}} \cot \left(\frac{x}{2^{n-1}} \right) = 2 \cot 2x.$$

14.

$$(i) \quad \tan A + \tan B + \tan(A+B) \tan A \tan B = \tan(A+B)$$

$$(ii) \quad \tan A - \tan B - \tan(A-B) \tan A \tan B = \tan(A-B)$$

$$(iii) \quad \tan A \tan B \tan C = \tan C - \tan A - \tan B \quad (\text{For } A+B=C)$$

15.

$$\tan(45+\theta) = \frac{1+\tan\theta}{1-\tan\theta}$$

$$\tan(45-\theta) = \frac{1-\tan\theta}{1+\tan\theta}$$

16.

$$\left. \begin{aligned} \sin^3 \theta &= 3 \sin \theta - 4 \sin^3 \theta \\ \cos^3 \theta &= 4 \cos^3 \theta - 3 \cos \theta \end{aligned} \right\}$$

$$\sin^3 \theta = \frac{3 \sin \theta}{4} - \frac{1}{4} \sin 3\theta$$

$$\cos^3 \theta = \frac{3 \cos \theta}{4} + \frac{1}{4} \cos 3\theta$$